CONVERSION OF NUMBER SYSTEM

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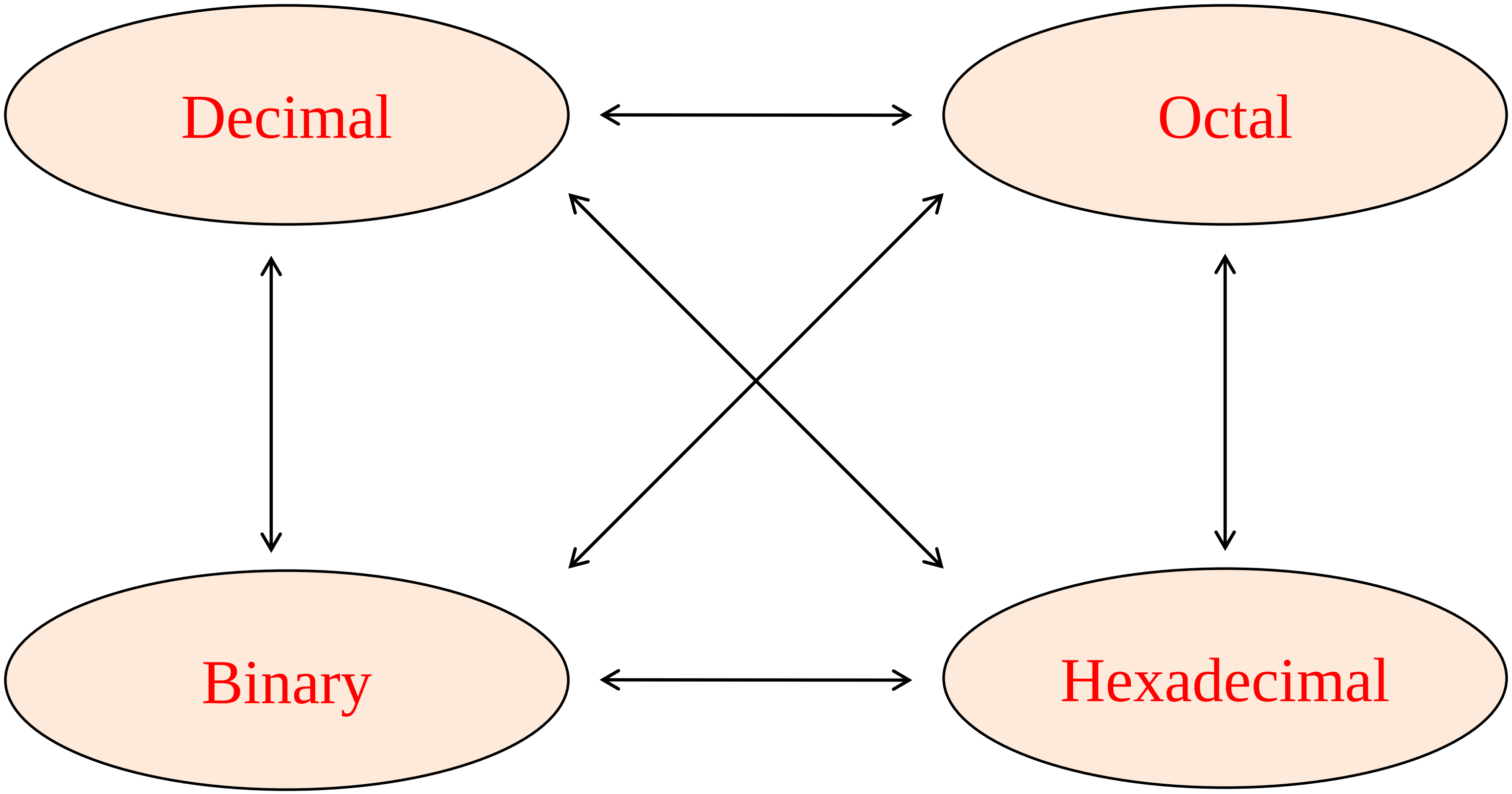
CONVERSION OF NUMBER SYSTEM

REFERENCE BOOKS

- Digital Systems, Principle & Applications, Ronald J. Tocci, Prentice-Hall
- Digital Design, M. Morris Mano, Michael D. Ciletti, Pearson Education, Inc.
- Digital Circuits and Design, S. Salivahanan, S. Arivazhagan, Oxford University Press.
- Digital Electronics, G. K. Kharate, Oxford University Press.
- Digital Electronics, Bignell James, Logic and Systems, Cengage Learning
- Digital Logic and Computer Design, M. Morris Mano, Pearson Education, Inc.

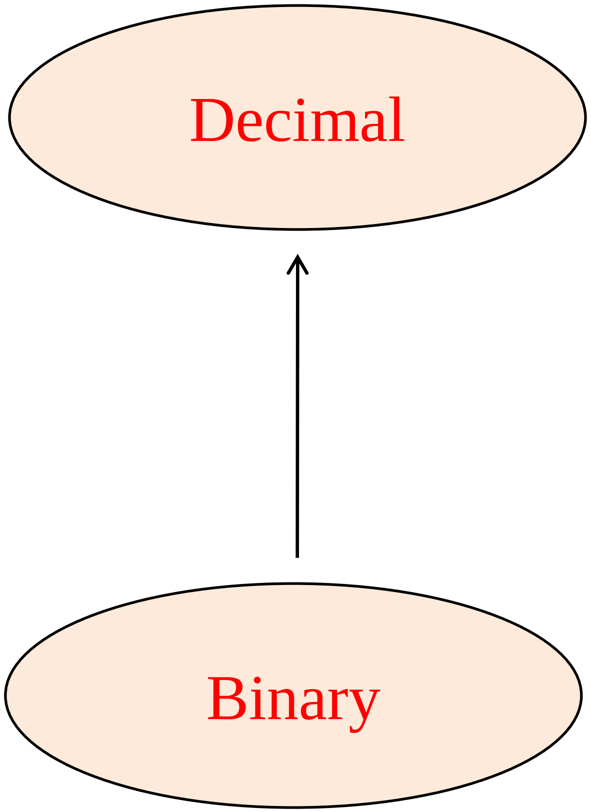
CONVERSION OF NUMBER SYSTEM

The possible conversions are



CONVERSION OF NUMBER SYSTEM

BINARY TO DECIMAL CONVERSION:



CONVERSION OF NUMBER SYSTEM

PROCEDURE:

- Multiply each digit by its weight i.e. by 2^n, where 'n' is the weight.
- The weight is the position of the digit, starting from 0 on the right side.
- Add all products to obtain the value of the number.

For example: $(10101)_2 = (?)_{10}$

BASE

$$(10101)_2 = 1\times 2^4 + 0\times 2^3 + 1\times 2^2 + 0\times 2^1 + 1\times 2^0$$

$$= 16 + 0 + 4 + 0 + 1$$

$$= 21$$

$$(10101)_2 = (21)_{10}$$

CONVERSION OF NUMBER SYSTEM

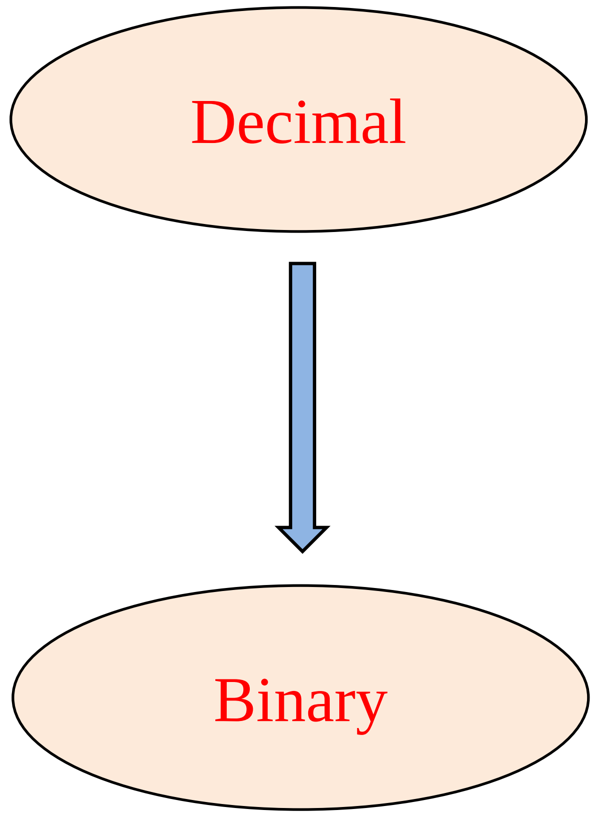
REPRESENTATION OF FRACTIONAL PART OF BINARY NUMBERS:

EXAMPLE:

$$
\begin{aligned}
(0.0111)_2 &= 0\times 2^{-1} + 1\times 2^{-2} + 1\times 2^{-3} + 1\times 2^{-4} \\
&= 0\times 1/2 + 1\times 1/4 + 1\times 1/8 + 1\times 1/16 \\
&= 0 + 1/4 + 1/8 + 1/16 \\
&= 0 + 0.25 + 0.125 + 0.0625 \\
&= (0.4375)_{10}
\end{aligned}
$$

CONVERSION OF NUMBER SYSTEM

DECIMAL TO BINARY CONVERSION:



CONVERSION OF NUMBER SYSTEM

PROCEDURE:

- For converting decimal to binary, repeated division method is used.
- Decimal number is divided successively by 2 till the quotient becomes zero.
- If the remainder is 0, on the right side write down a 0.
- If the remainder is 1, write down a 1.
- The LSB is of binary number is the first remainder and MSB is the last remainder.

CONVERSION OF NUMBER SYSTEM

EXAMPLE: convert $(50)_{10}$ into binary.

Division		Remainder	
2	50	0	LSB
2	25	1	
2	12	0	
2	6	0	
2	3	1	
2	1	1	
2	0	1	MSB

List all remaiders in reverse order (bottom to top) will give equivalent binary.

$$(50)_{10} = (110010)_2$$

CONVERSION OF NUMBER SYSTEM

CONVERSION FROM DECIMAL FRACTION TO BINARY:

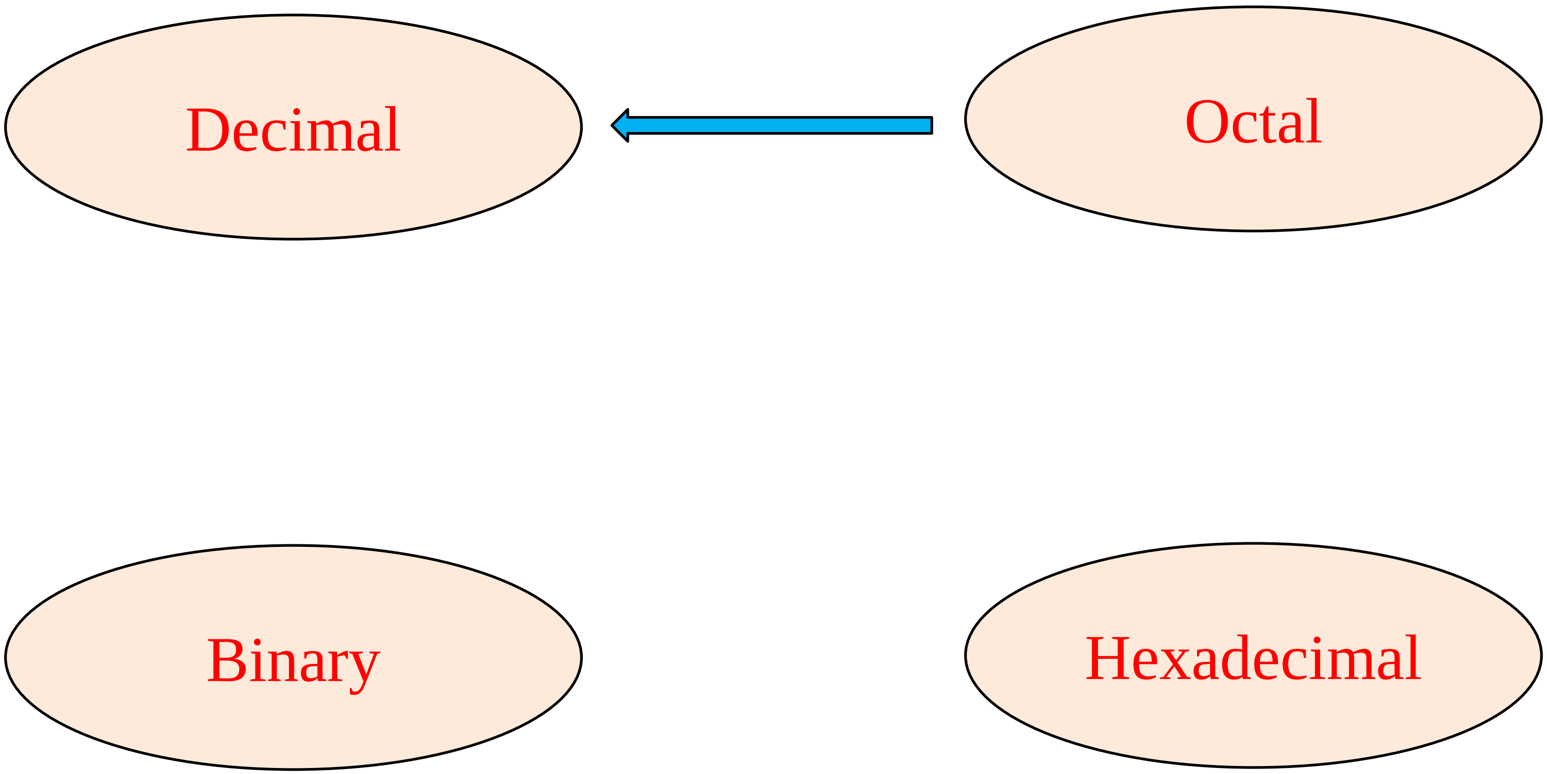
- To convert a decimal fraction to binary, multiply the decimal fraction successively by 2, till the fractional part of the product is zero.
- List all the integer part of the multiplication, reading from top to bottom, to give the binary equivalent of the decimal number. For example

Multiplication			Integer	
0.625 * 2	=	1.25	1	MSB
0.25 * 2	=	0.50	0	
0.50 * 2	=	1.00	1	
0.00 * 2	=	0.00	0	LSB

$$(0.625)_{10} = (0.101)_2$$

CONVERSION OF NUMBER SYSTEM

OCTAL TO DECIMAL CONVERSION:



CONVERSION OF NUMBER SYSTEM

PROCEDURE:

- ❖Multiply each digit by its weight i.e. by 8^n, where ‘n’ is the weight.
- ❖ The weight is the position of the digit, starting from 0 on the right side.
- ❖ Add all products to obtain the value of the number.
- ❖The result will be the equivalent decimal number

EXAMPLE: Convert $(431.510)_8$ into decimal number.

$$(431.510)_8 = 4\times 8^2 + 3\times 8^1 + 1\times 8^0 + 5\times 8^{-1} + 1\times 8^{-2} + 0\times 8^{-3}$$

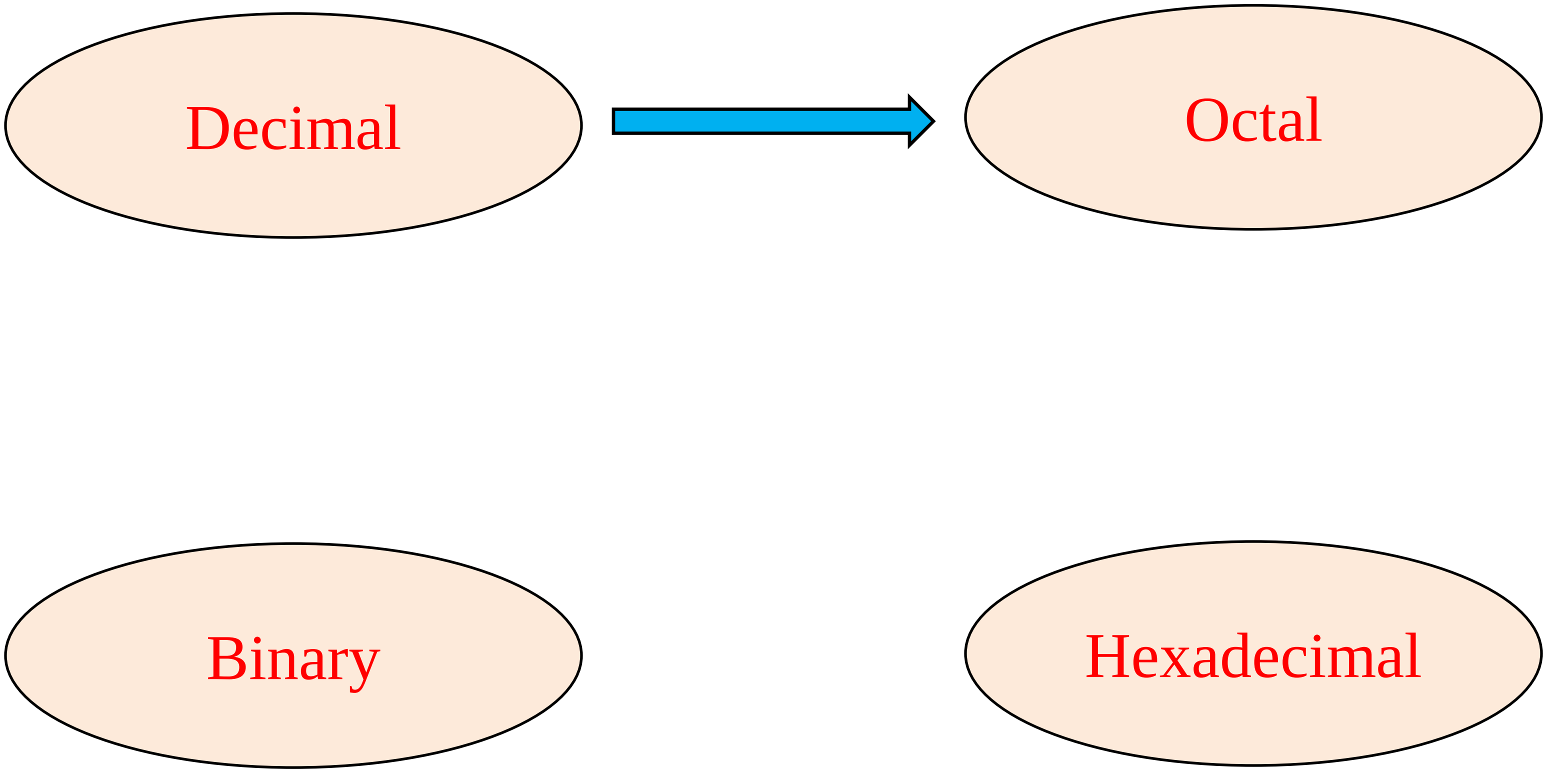
$$= 256 + 24 + 1 + 0.625 + 0.015625$$

$$= 281.640625$$

$$\therefore (431.510)_8 = (281.640625)_{10}$$

CONVERSION OF NUMBER SYSTEM

DECIMAL TO OCTAL CONVERSION:



CONVERSION OF NUMBER SYSTEM

PROCEDURE:

- For converting decimal to octal, repeated division method is used.
- Decimal number is divided successively by 8 till the quotient becomes zero.
- If the remainder is 0, on the right side write down a 0.
- If the remainder is 1, write down a 1.
- The LSB is of octal number is the first remainder and MSB is the last remainder.

CONVERSION OF NUMBER SYSTEM

EXAMPLE: convert $(150)_{10}$ into octal number.

Division		Remainder
8	150	6
8	18	2
8	2	2
	0	

LSB

MSB

List all the remainders in reverse order (bottom to top) will give equivalent octal number.

$(150)_{10} = (226)_8$

CONVERSION OF NUMBER SYSTEM

CONVERSION FROM DECIMAL FRACTION TO OCTAL:

PROCEDURE:

- To convert a decimal fraction to octal, multiply the decimal fraction successively by 8, till the fractional part of the product is zero.
- List all the integer part of the multiplication, reading from top to bottom, to give the octal equivalent of the decimal fraction. For example:

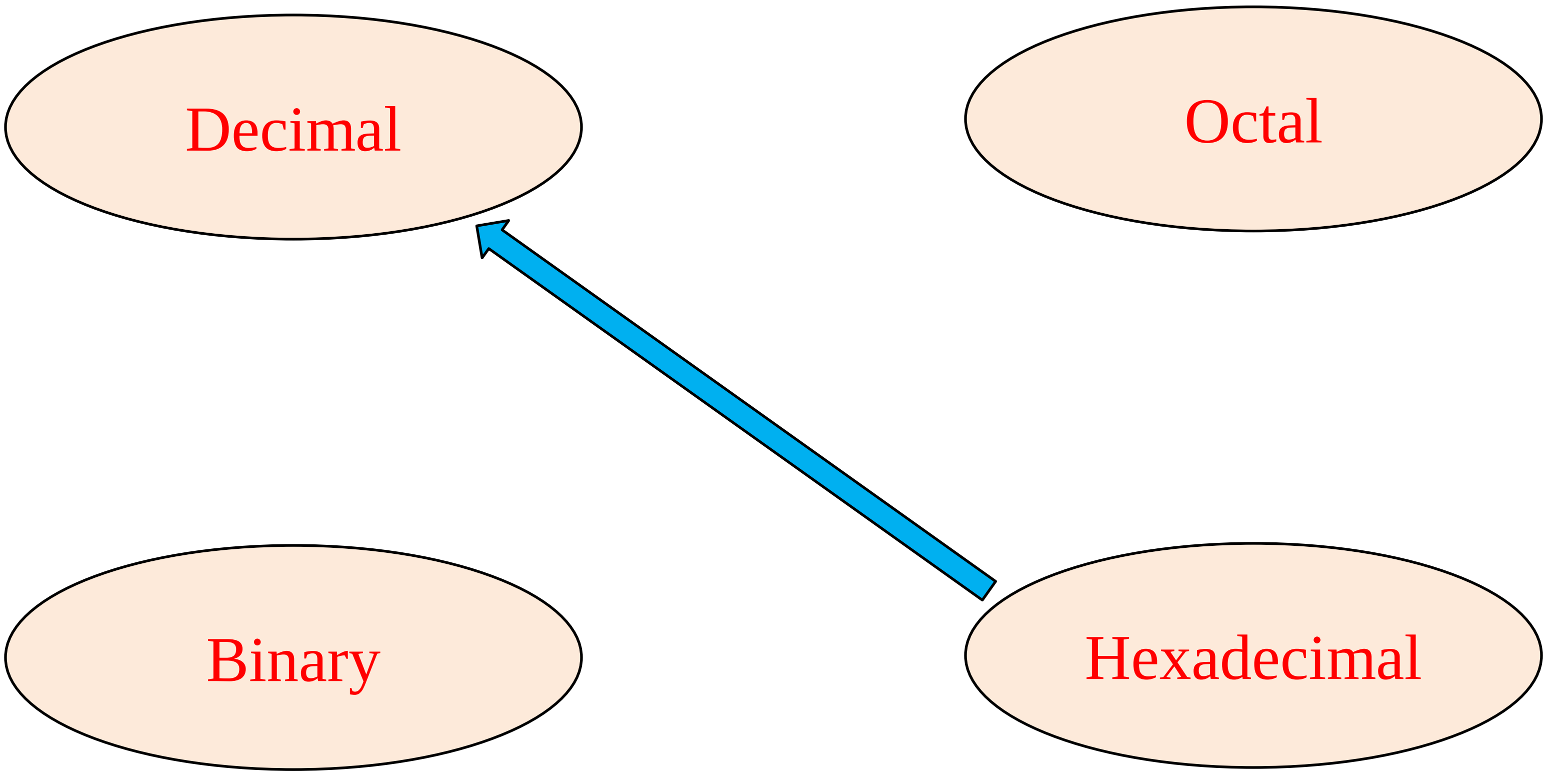
Convert $(0.456)_{10} = (?)_8$

Multiplication	Integer
$0.456 * 8 = 3.648$	3
$0.648 * 8 = 5.184$	5
$0.184 * 8 = 1.472$	1
$0.472 * 8 = 3.776$	3
$0.776 * 8 = 6.208$	6

$(0.456)_{10} = (0.35136)_8$

CONVERSION OF NUMBER SYSTEM

HEXADECIMAL TO DECIMAL CONVERSION:



CONVERSION OF NUMBER SYSTEM

PROCEDURE:

- Multiply each digit by its weight i.e. by 16^n, where ‘n’ is the weight.
- The weight is the position of the digit, starting from 0 on the right side.
- Add all products to obtain the value of the number.
- The result will be the equivalent decimal number

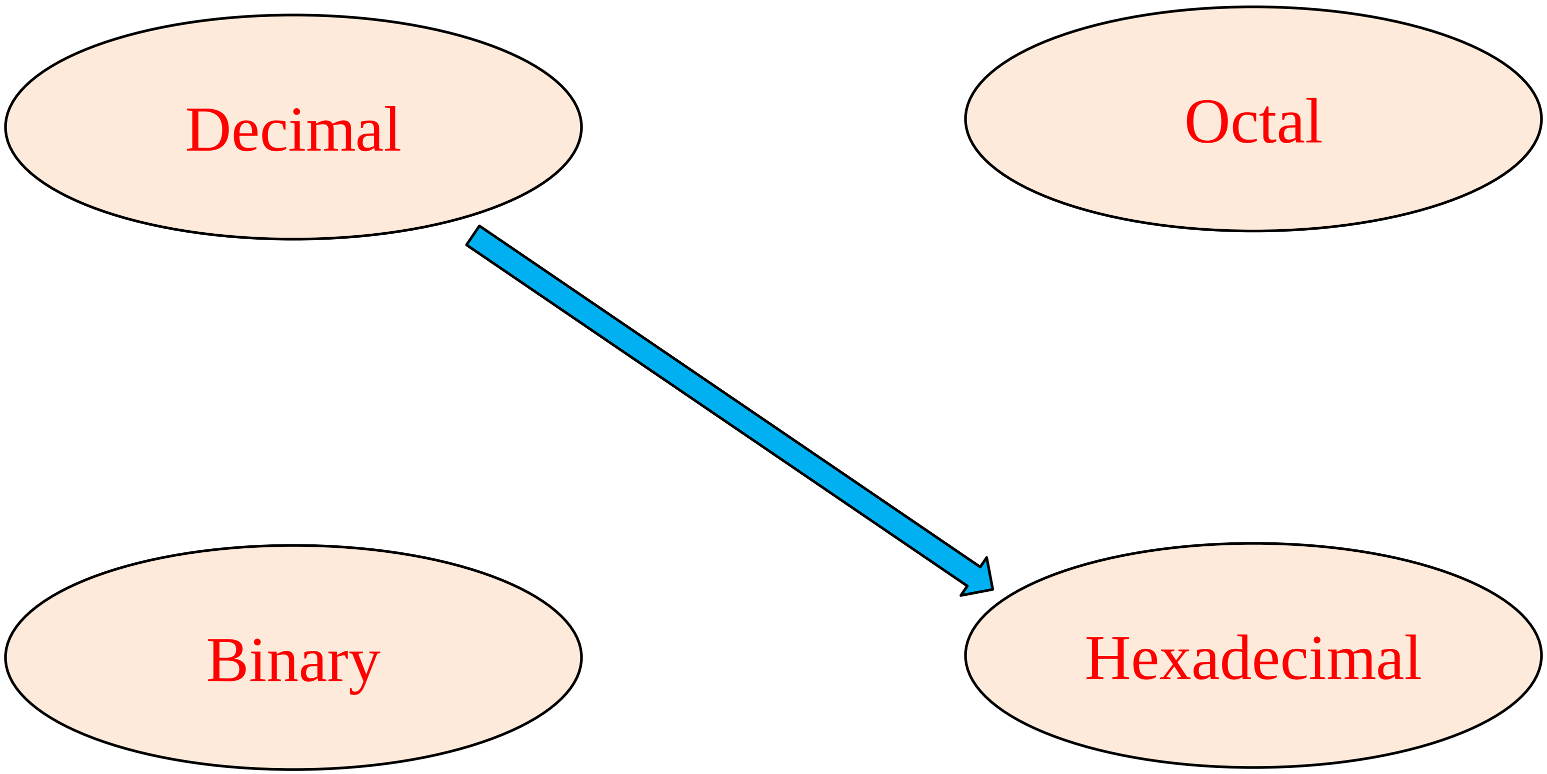
EXAMPLE: Convert $(AB6.C)_{16}$ into decimal number

$$
\begin{aligned}
(AB6.C)_{16} &= A\times16^{2} + B\times16^{1} + 6\times16^{0} + C\times16^{-1} \\
&= 10\times256 + 11\times16 + 6\times1 + 12\times0.0625 \\
&= 2560 + 176 + 6 + 0.75 \\
&= 2742.75
\end{aligned}
$$

$$(AB6.C)_{16} = (2742.75)_{10}$$

CONVERSION OF NUMBER SYSTEM

DECIMAL TO HEXADECIMAL CONVERSION:



CONVERSION OF NUMBER SYSTEM

PROCEDURE:

- For converting decimal to hexadecimal, repeated division method is used.
- Decimal number is divided successively by 16 till the quotient becomes zero.
- If the remainder is 0, on the right side write down a 0.
- If the remainder is 1, write down a 1.
- The LSB is of hexadecimal number is the first remainder and MSB is the last remainder.

CONVERSION OF NUMBER SYSTEM

EXAMPLE:

Convert $(450)_{10} = (?)_{16}$

Division		Remainders
16	450	2
16	28	12 = C
16	1	1
16	0	

LSD

MSD

$(450)_{10} = (1C2)_{16}$

CONVERSION OF NUMBER SYSTEM

CONVERSION FROM DECIMAL FRACTION TO HEXADECIMAL:

PROCEDURE:

- To convert a decimal fraction to hexadecimal, multiply the decimal fraction successively by 16, till the fractional part of the product is zero.
- List all the integer part of the multiplication, reading from top to bottom, to give the hexadecimal equivalent of the decimal fraction. For example:

Multiplication	Integer Part
$0.03125 * 16 = 0.5$	0
$0.5 * 16 = 8.0$	8

Therefore, $(0.03125)_{10} = (0.08)_{16}$

THANK YOU

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